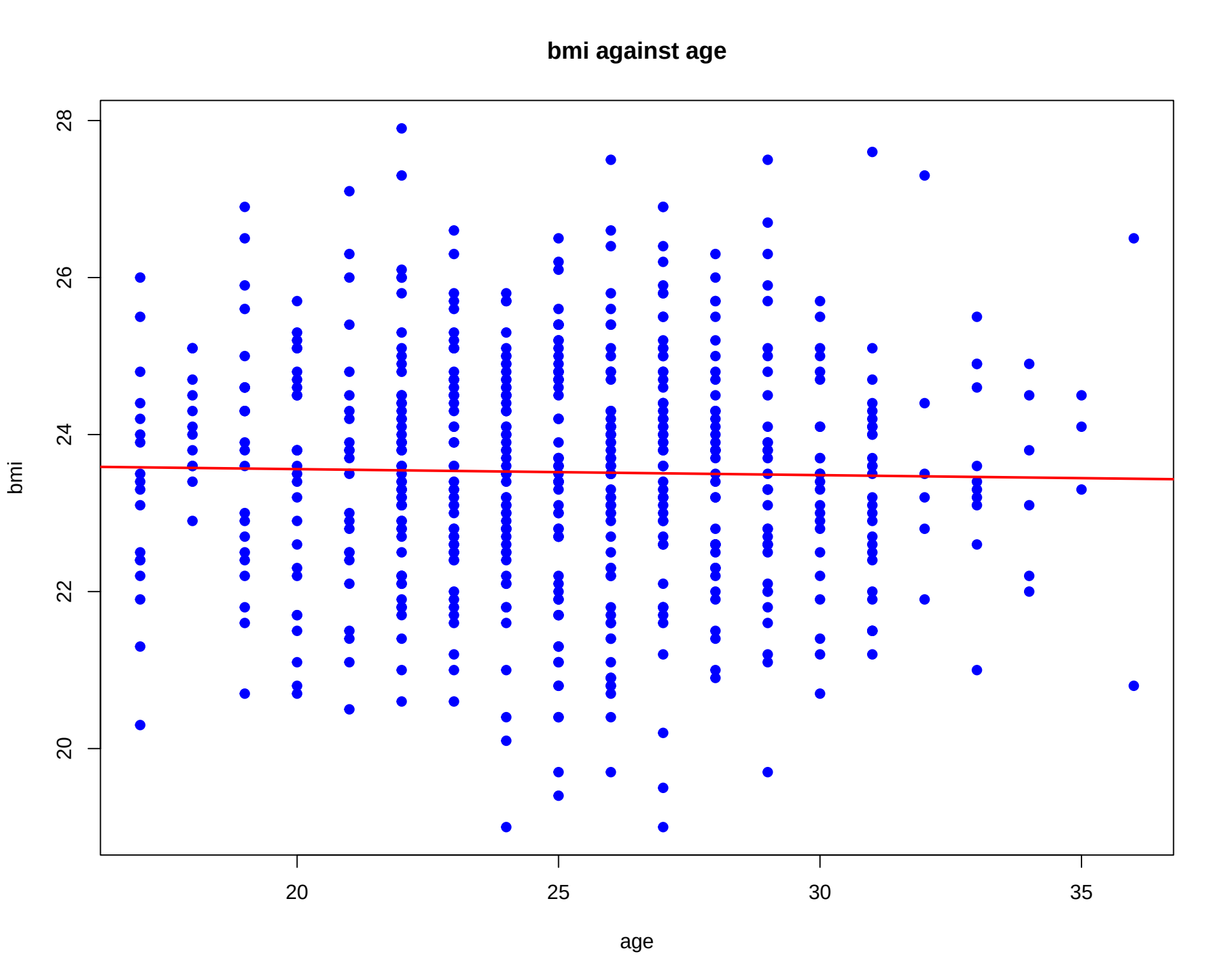
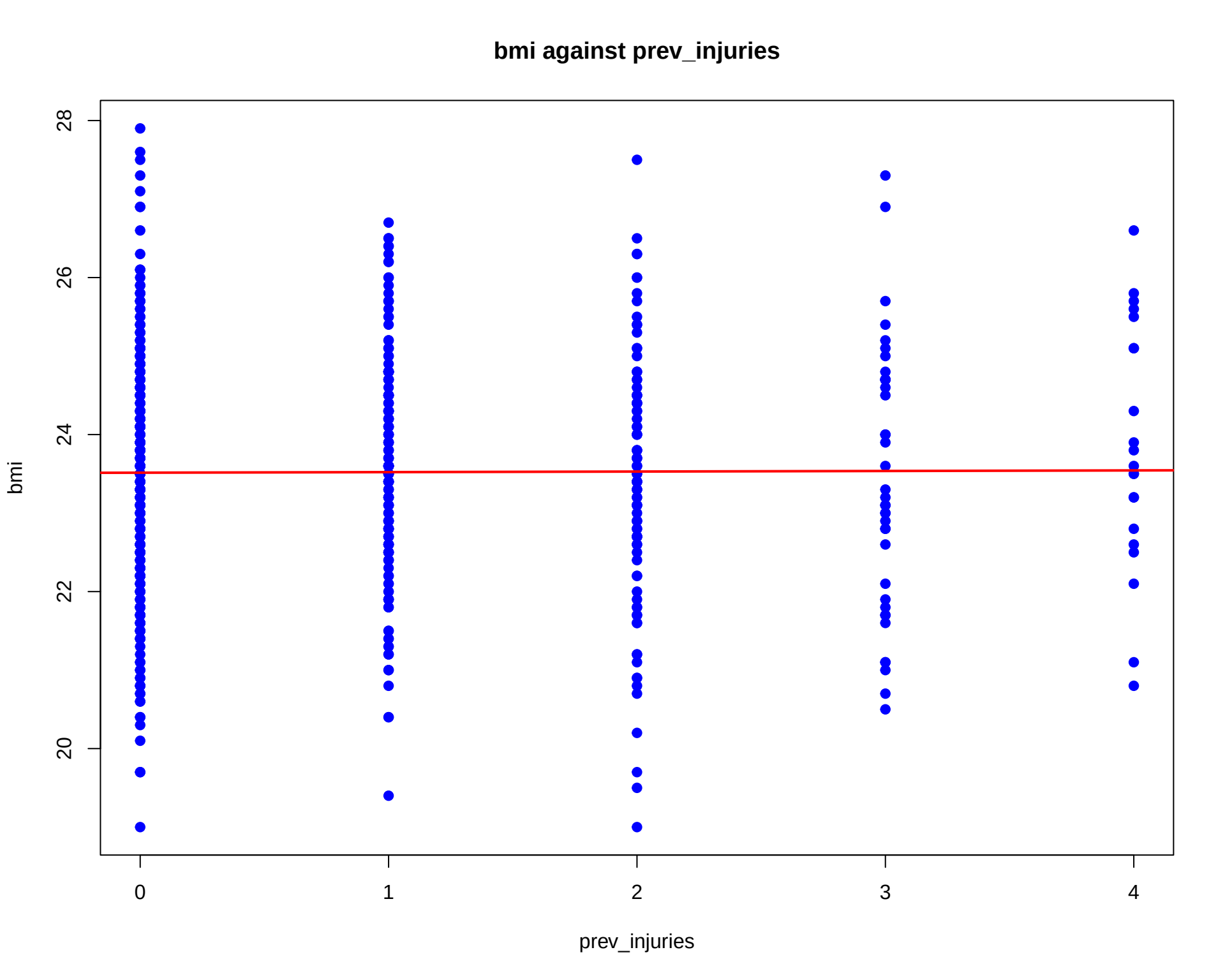
bmi ~ age+prev_injuries+injured

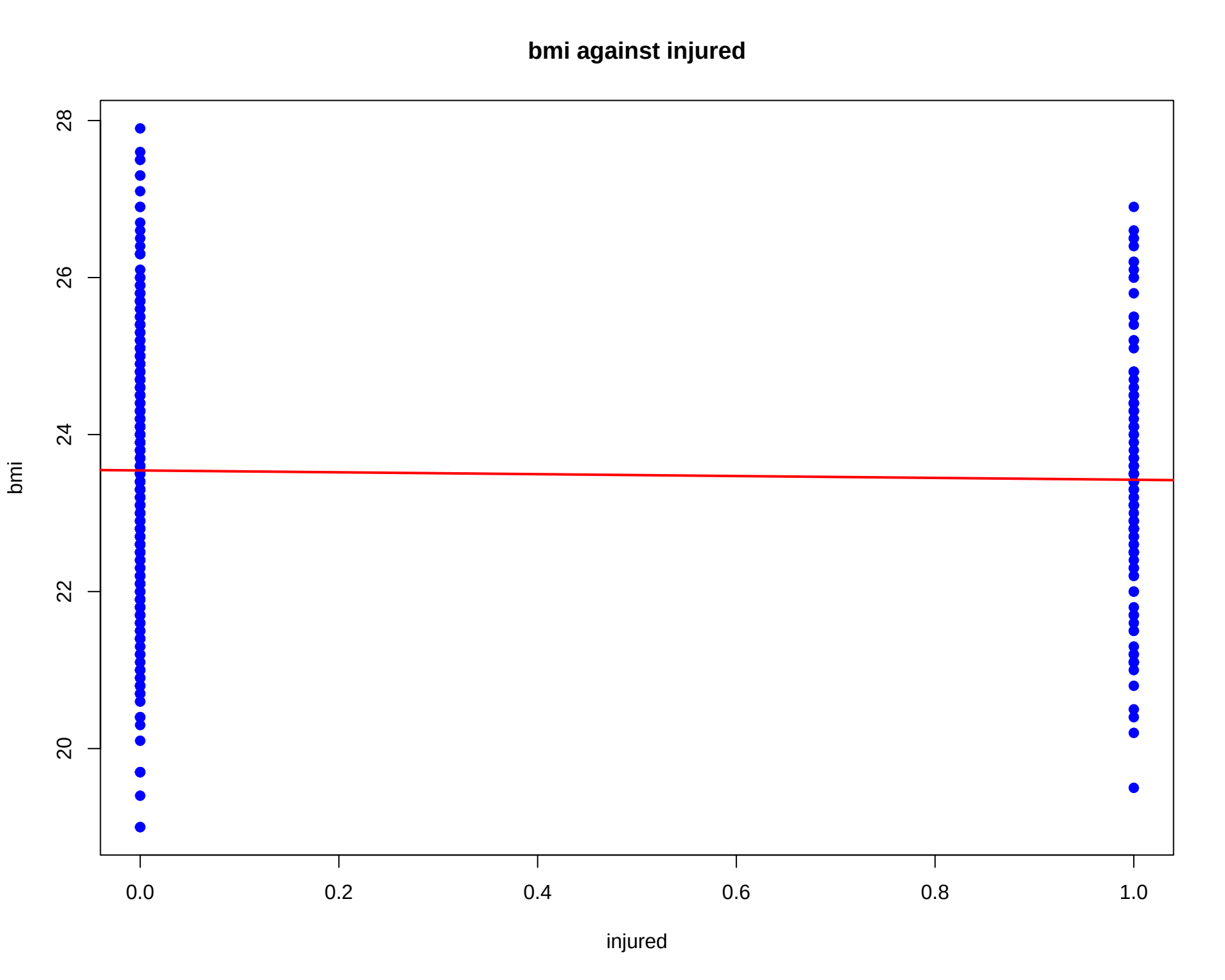
23.675 + (−0.006)*age + (0.018)*prev_injuries + (−0.122)*injured

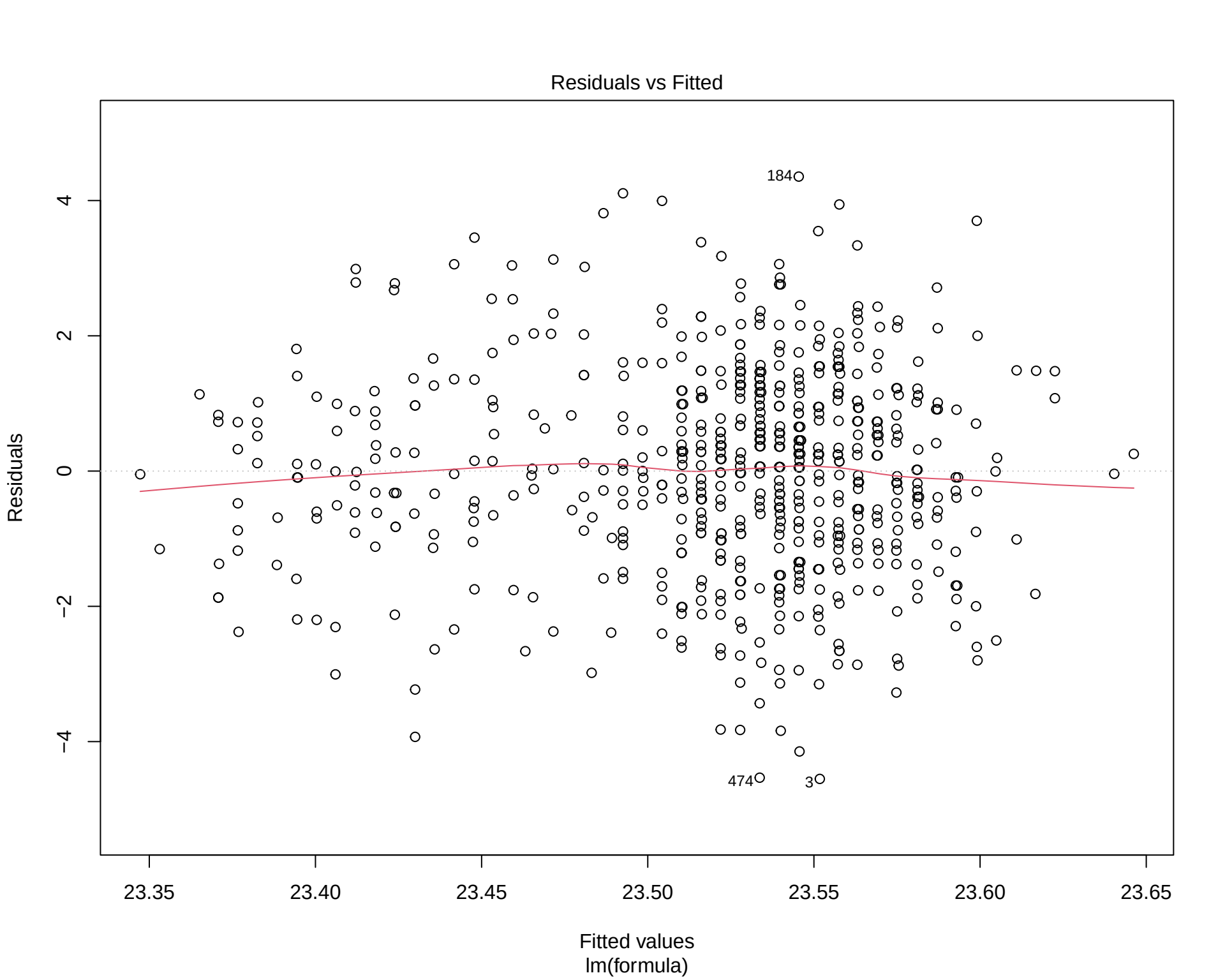
A good model will have: big F−stat (at least >3 depending on n, 10+ is nice), small p−value < 0.05,
High R and Adj R,
Low AIC

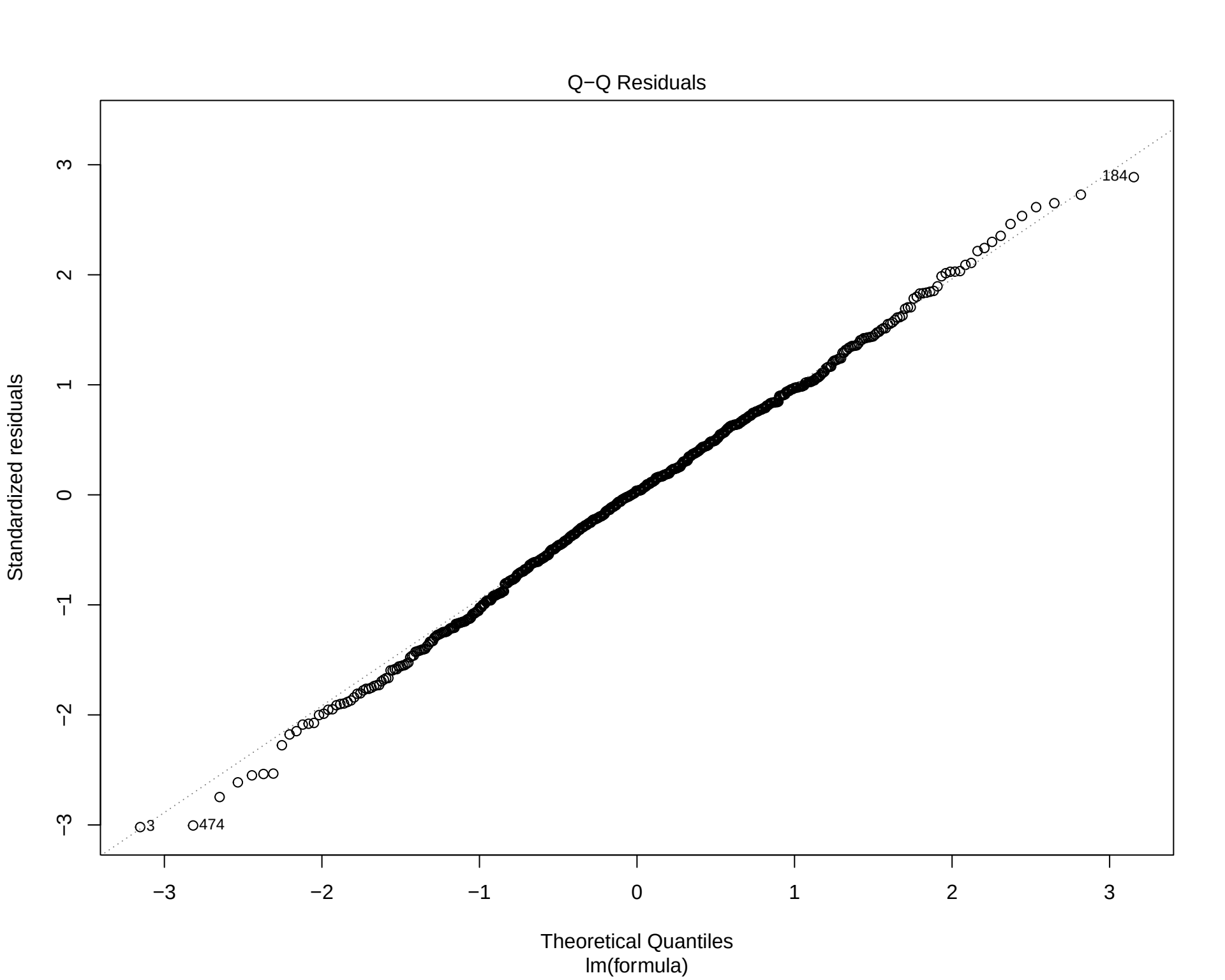
F−statistic: 0.27197 on 3 and 616 Degrees of Freedom
F p−value: 0.84563
R−Squared: 0.00132
Adjusted R−Squared: −0.00354091847900628
Root MSE: 1.50606555284935
AIC: 2277.26459488511

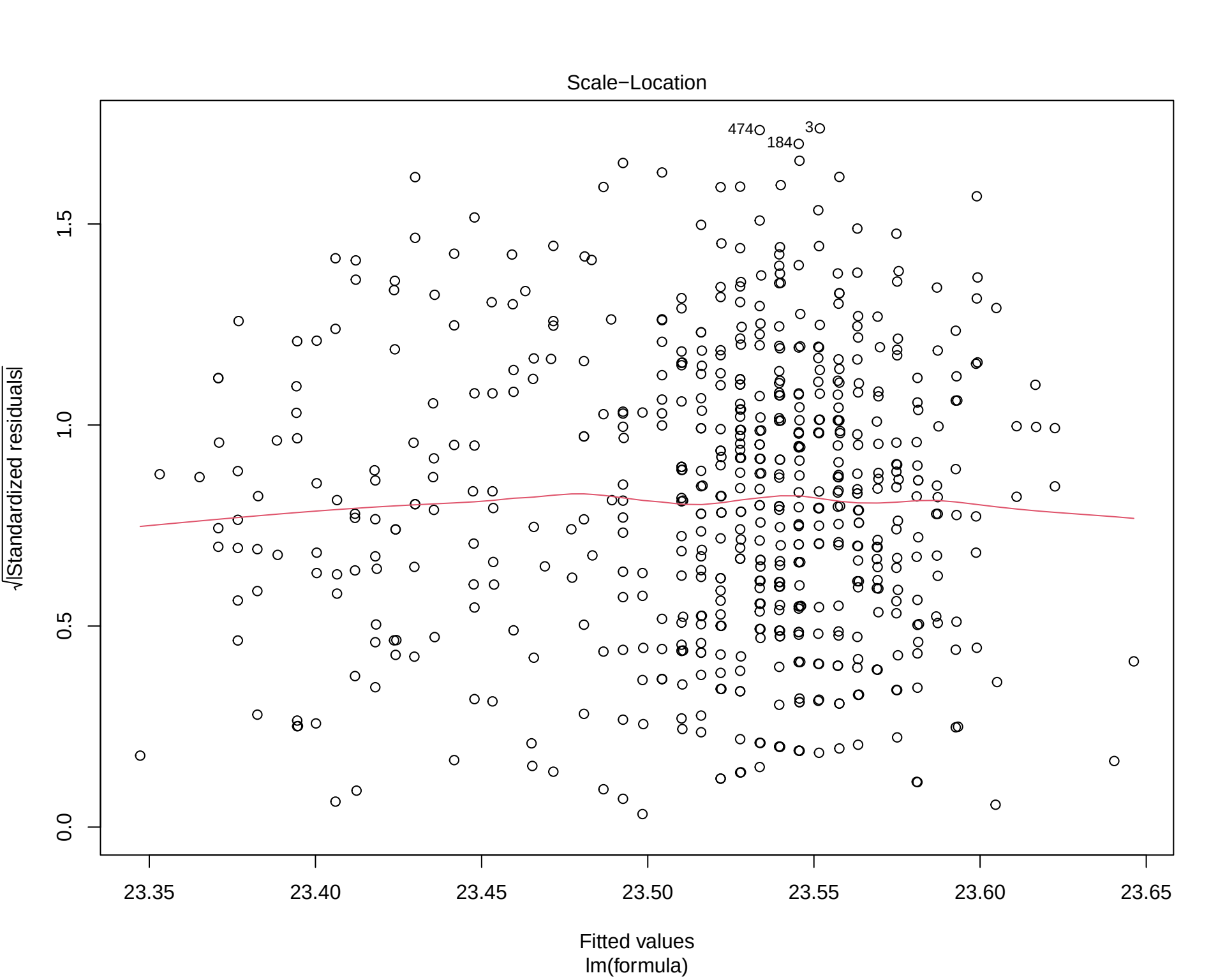

bmi against age
bmi
age
20
22
24
26
28
20
25
30
35

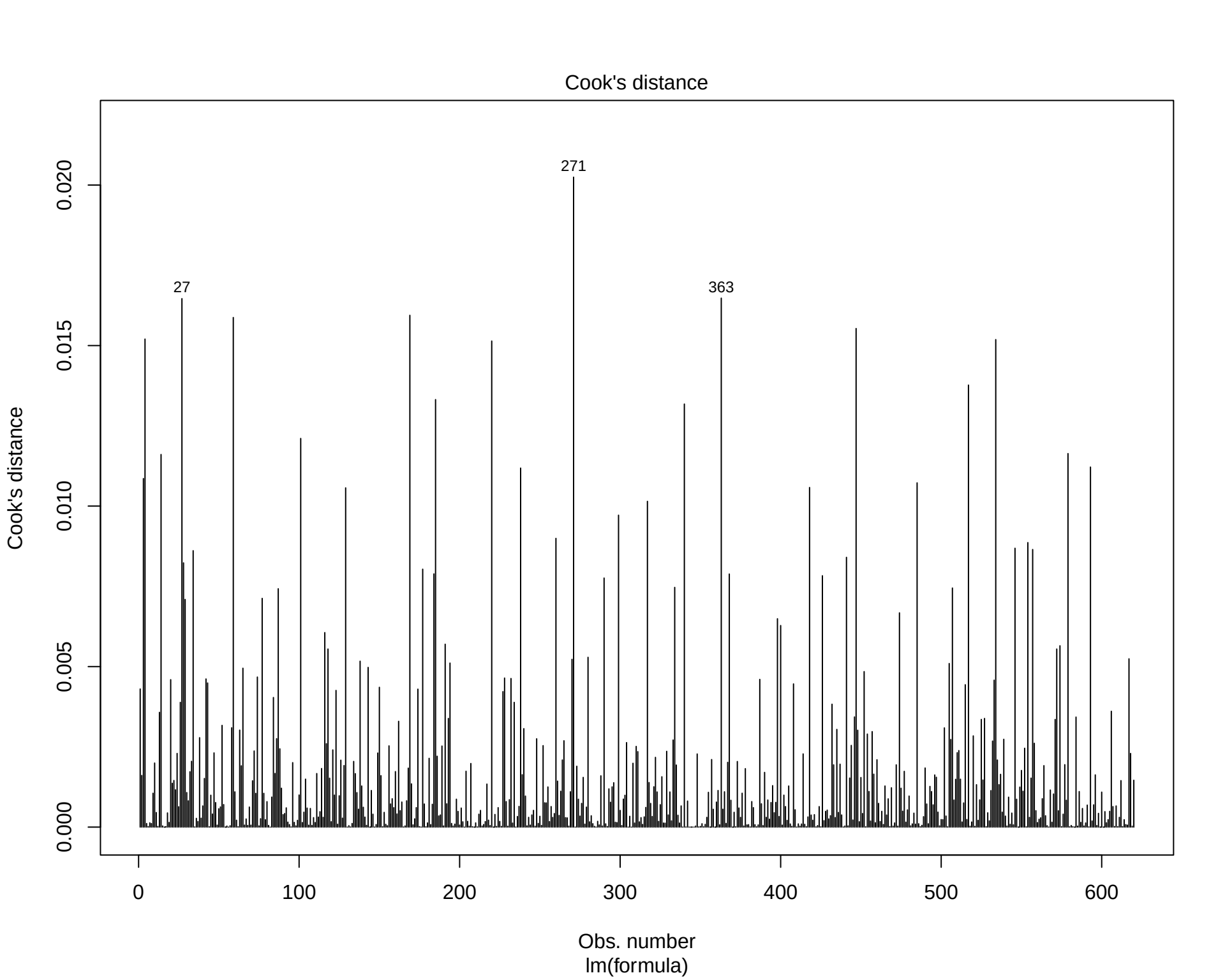

bmi against prev_injuries
bmi
prev_injuries
20
22
24
26
28
0
1
2
3
4

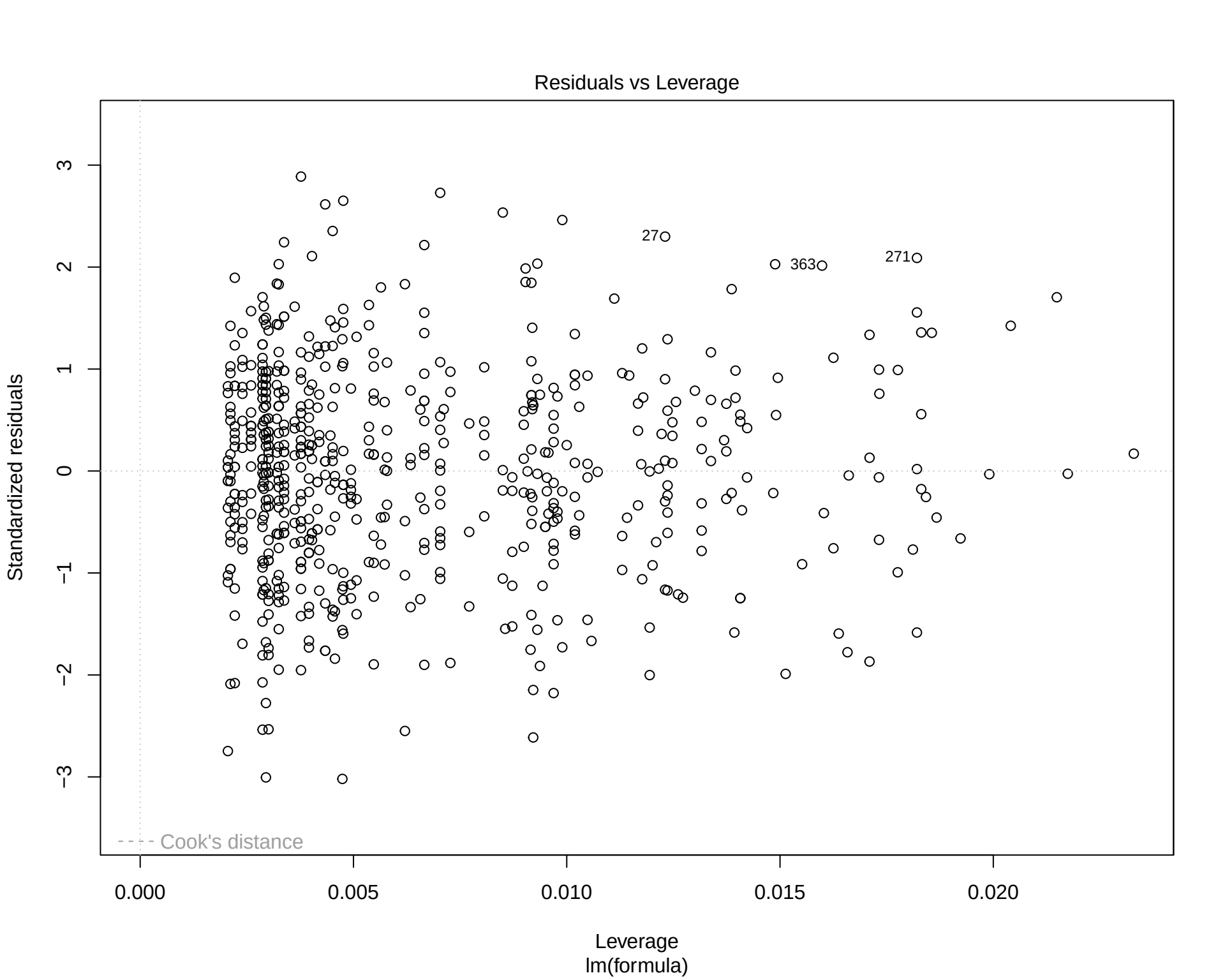

bmi against injured
bmi
injured
20
22
24
26
28
0.0
0.2
0.4
0.6
0.8
1.0

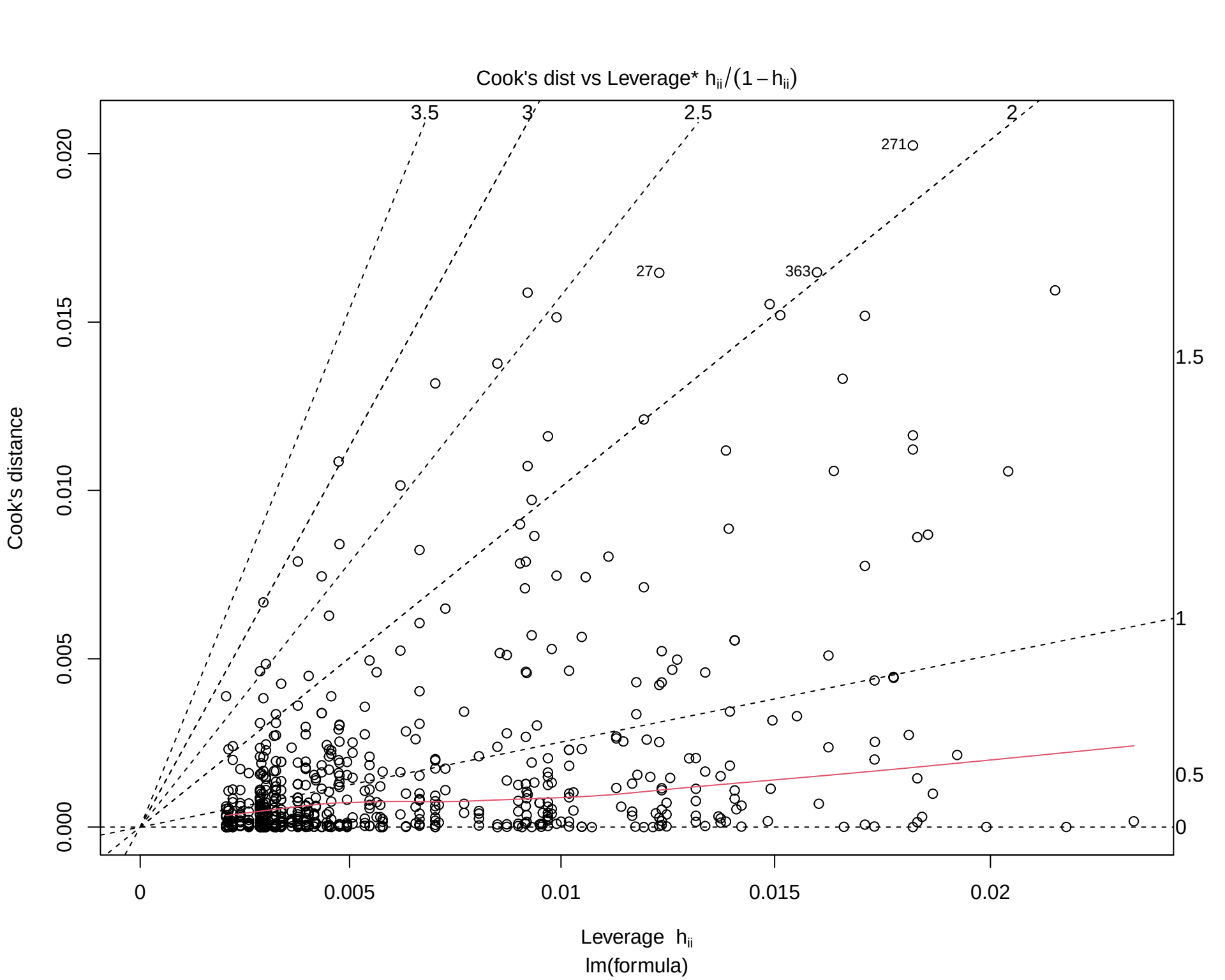

Residuals vs Fitted
Residuals
Fitted values
lm(formula)
184
474
3
4
2
0
−2
−4
23.35
23.40
23.45
23.50
23.55
23.60
23.65


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(formula)
3
474
184
−3
−2
−1
0
1
2
3


Scale−Location
474
184
3
√|Standardized residuals|
0.0
0.5
1.0
1.5
23.35
23.40
23.45
23.50
23.55
23.60
23.65
Fitted values
lm(formula)


Cook's distance
271
27
363
0.020
0.015
0.010
0.005
0.000
Cook's distance
0
100
200
300
400
500
600
Obs. number
lm(formula)


Residuals vs Leverage
Standardized residuals
3
2
1
0
−1
−2
−3
27
363
271
Cook's distance
0.000
0.005
0.010
0.015
0.020
Leverage
lm(formula)


Cook's dist vs Leverage* $h_{ii}/(1-h_{ii})$
3.5
3
2.5
2
271
27
363
Cook's distance
0.000
0.005
0.010
0.015
0.020
0
0.5
1
1.5
0
0.005
0.01
0.015
0.02
Leverage h_{ii}
lm(formula)